

UNIVERSITY OF TABUK

Faculty of Computers & Information Technology
Department of Computer Engineering

Course: CEN-431 - Computer Networks Lab

Course Project:
**Design and Implementation of a
Multi-Service Network Using Cisco
Packet Tracer**

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1 Introduction

This document serves as the comprehensive course project report for the CEN-431 Computer Networks Lab. The primary objective of this project was to design, configure, and validate a scalable enterprise network for a fictional organization using the Cisco Packet Tracer simulation environment. The implementation successfully integrates wired and wireless connectivity, deploying dynamic routing protocols—specifically Routing Information Protocol (RIP) and Enhanced Interior Gateway Routing Protocol (EIGRP)—across multiple sites. Furthermore, the architecture incorporates essential network services including Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), and Network Address Translation (NAT) to ensure a robust, secure, and fully functional multi-service network environment.

2 Network Topology & Design

The overarching network architecture, detailed visually in Figure 1, is logically segmented into three primary areas to simulate a realistic corporate infrastructure:

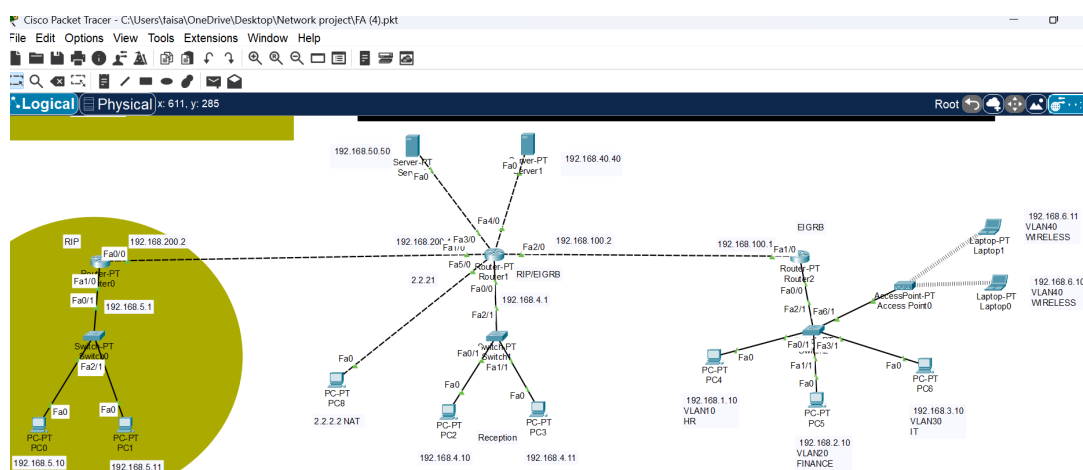


Figure 1: Course Project Network Topology Diagram: Main Office, Core/WAN, and Branch Office Logical Design

- **Main Office (EIGRP Domain):** Represents the corporate headquarters, utilizing EIGRP for rapid convergence and efficient internal routing.
 - *Hardware Inventory:* 2 Routers, 2 Layer 2 Switches, 1 Wireless Access Point, 4 PCs, 2 Laptops, 1 DHCP Server, and 1 DNS Server.
- **Branch Office (RIP Domain):** Represents a remote location utilizing RIP for its internal routing domain.
 - *Hardware Inventory:* 1 Router, 1 Layer 2 Switch, and 2 PCs.

- **Core/WAN:** The interconnecting Wide Area Network links bridging the Main Office, the Branch Office, and the simulated external internet.

3 Switching and VLAN Configuration

To optimize broadcast domains, enhance security, and logically group users by department, Virtual Local Area Networks (VLANs) were implemented across the Layer 2 switches in the Main Office.

- **VLAN Segmentation:** The network is divided into four primary VLANs:
 - **VLAN 10:** HR (Human Resources)
 - **VLAN 20:** Finance
 - **VLAN 30:** IT
 - **VLAN 40:** Wireless Devices
- **Port Configuration:** Switch interfaces were explicitly configured as access ports and mapped to their respective VLANs (e.g., interface FastEthernet 1/1 was assigned to VLAN 20, and FastEthernet 6/1 to VLAN 40).
- **Trunking:** To allow traffic from multiple VLANs to traverse a single physical link between the switches and the gateway router, IEEE 802.1Q trunking was established, notably on interface FastEthernet 2/1 using the `switchport mode trunk` command.

4 Routing Protocols & Inter-VLAN Routing

A multi-protocol routing environment was established to facilitate communication between isolated subnets and distinct autonomous systems.

- **Router-on-a-Stick:** Inter-VLAN routing was achieved using the "Router-on-a-Stick" topology. Virtual subinterfaces (e.g., 0/0.10, 0/0.20, 0/0.30) were created on the Main Office router's physical interface. Each subinterface was configured with `encapsulation dot1Q` to tag traffic appropriately and assigned the default gateway IP address for its respective VLAN.
- **EIGRP Configuration:** EIGRP was configured in the Main Office using Autonomous System (AS) 10 to manage routing between the internal subnets and the core network.
- **RIP Configuration:** RIP version 2 was deployed on the Branch Office router and the connecting WAN link.

- **Route Redistribution:** To enable communication between the EIGRP and RIP domains, mutual route redistribution was performed at the boundary router. Because EIGRP relies on a composite metric, RIP routes were redistributed into EIGRP using custom seed metrics. The critical configuration applied was: `redistribute rip metric 1000 1 255 1 1500` (Bandwidth: 1000, Delay: 1, Reliability: 255, Load: 1, MTU: 1500).

5 Network Services (DHCP, NAT, Wireless)

Crucial application-layer services were deployed to automate network management, secure access, and provide external connectivity.

- **DHCP Implementation:** A centralized DHCP Server (IP: 192.168.50.50) was configured with multiple IP pools tailored for each VLAN. Because routers block broadcast traffic by default, the `ip helper-address 192.168.50.50` command was applied to the router's subinterfaces. This allowed DHCP DISCOVER broadcasts from clients to be converted into unicast packets and forwarded to the central server.
- **Wireless Security:** A Wireless Access Point was deployed to service the laptops in VLAN 40. It was configured with the SSID "**Wireless**" on Channel 6, utilizing WPA2-PSK (Pre-Shared Key) security with AES encryption to ensure data confidentiality.
- **NAT & Internet Connectivity:** A Loopback interface with the IP address 1.1.1.1 was created on the edge router to simulate the public Internet. Network Address Translation (NAT) Overload, also known as Port Address Translation (PAT), was configured to allow internal hosts to access this external network. Access Control List (ACL) 1 (`access-list 1 permit any`) was defined to identify permitted internal traffic.

6 Testing and Verification Procedures

Rigorous testing was conducted to validate the integrity and functionality of the network:

- **Inter-VLAN Connectivity:** ICMP Echo Requests (pings) were successfully transmitted between devices residing in different VLANs, verifying the Router-on-a-Stick configuration.
- **DHCP Allocation Verification:** Commands such as `ipconfig /renew` were executed on end devices, confirming that PCs and Laptops dynamically received the correct IP addresses, subnet masks, default gateways, and DNS server addresses.

- **End-to-End Routing:** Pings initiated from the Branch Office PCs successfully reached the Main Office servers and the simulated Internet loopback interface (1.1.1.1), validating the RIP/EIGRP redistribution and NAT Overload configurations.

7 Challenges Encountered & Resolutions

During the configuration phase, several technical obstacles were identified and resolved:

- **Challenge 1: DHCP Broadcast Drops Across Subnets**

Issue: End devices in VLANs 10, 20, and 30 were unable to acquire IP addresses dynamically, as their broadcast requests could not pass the local router interface.

Resolution: The issue was remediated by implementing the `ip helper-address 192.168.50.50` command on the respective subinterfaces of the gateway router, successfully relaying the requests to the centralized DHCP server.

- **Challenge 2: Route Redistribution Failure**

Issue: Initially, the Main Office devices could not reach the Branch Office network. While the `redistribute rip` command was entered under the EIGRP routing process, the routing tables did not populate.

Resolution: EIGRP requires explicit seed metrics to understand routes from other protocols. By appending the specific metrics to the redistribution command (`metric 1000 1 255 1 1500`), the router successfully calculated the feasible distance, and end-to-end connectivity was established.

8 Conclusion

The multi-service network designed for this project was successfully implemented and meets all the specified requirements. By effectively segmenting traffic via VLANs, integrating dynamic routing protocols with mutual redistribution, and deploying essential network services like DHCP, PAT, and secure wireless access, the resulting architecture represents a highly functional, secure, and scalable enterprise network simulation.